

APPENDIX B
FOR
PALLET RACK
AT
PITTSBURGH ANGB, PENNSYLVANIA
BUILDING: 110
CSN NPT501

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1. GENERAL SCOPE

This appendix establishes the minimum performance, design, fabrication, installation, and test requirements for pallet rack types A, B, C and D at Pittsburgh ANGB, Building 110. The work shall be completed in accordance with this appendix, the Cover Purchase Description (PD), and Air Force (AF) Drawing NPT501-01. This appendix covers all equipment, materials, and labor necessary to complete installation of the system.

2. APPLICABLE DOCUMENTS

Applicable documents shall be in accordance with Section 2 of the Cover PD as well as RMI/ANSI MH16.1 *Industrial Steel Storage Racks* and RMI/ANSI MH26.2 *Design and Testing of Welded Wire Rack Decking*.

3. REQUIREMENTS

3.1. SYSTEM DESCRIPTION

The manufacturer of the pallet rack and decking shall be R-Mark® certified.

3.2. CHARACTERISTICS

The system shall conform to the following requirements.

3.2.1. CONFIGURATION REQUIREMENTS

The pallet rack units shall be located and configured as shown on AF Drawing NPT501-01.

3.2.2. STRUCTURE AND INSTALLATION

- UPRIGHTS
 - Capacity as shown on AF drawing NPT501-01.
 - Capable of supporting fully loaded bays without permanent deformation.
 - Tear drop openings spaced on 2-inch increments for shelf beams.
- BEARING PLATES
 - Located at bottom of each column, able to sufficiently disperse load.
 - Minimum of 2 punched holes for ½ inch anchor bolts.
- SHELF BEAMS
 - Capacity as shown on AF Drawing NPT501-01.
 - Teardrop compatible connectors.
 - Attached shelf beam-to-column safety lock.
 - Step type.

- ROW SPACERS
 - 6-inch flue space as shown on AF Drawing NPT501-01.
 - 2 spacers provided within 24 inches from top and bottom.
- LEVELING SHIMS
 - Length and width equal to bearing plate.
 - No more than 5 shims used at any one location.
 - No slotted or welded shims.
- DECKING
 - R-Mark® compliant.
 - Galvanized.
 - External waterfall to interface with step beam.
 - Support beams permanently attached to decking.
- ANCHORING
 - Rigid freestanding structure.
 - Installed as per manufacturers instructions for the geographic location.
- ASSEMBLY
 - Be straight and square.
 - Horizontal members level to within 1/8 inch over 10 feet.
 - Vertical members plumb to within 1/2 inch over 10 feet.
- CAPACITY LABELS
 - Shall be 8.5 X 11 Inches.
- ROW MARKERS
 - Two card holders installed on the end of each rack section for row identification.
 - Card holders capable of holding and shielding 8.5 X 11 inch printed labels from debris.
 - Securely mounted between 7 feet and 10 feet from the floor.
- COLOR
 - Shall be as specified by COTR.

3.2.3. IN-RACK SPRINKLER SYSTEM

An in-rack sprinkler system is not required.

4. QUALITY ASSURANCE PROVISIONS

4.1. INSPECTION

The system shall be inspected to verify that it meets all the requirements of sections 1, 2, and 3 of this appendix and the Cover PD. These inspections shall be completed as part of the system checkout test and again as a part of the quality conformance test

4.2. TESTS

The following tests shall be performed on the system as part of the system checkout test and again as a part of the quality conformance test.

No.	Description	Result
1.	Maximum loads shall be placed randomly in the rack system. After not less than four hours, the test loads shall be removed and the rack examined for signs of deformation. If any component of the system exhibits plastic or elastic deformation outside of the approved parameters, the system fails the test.	☐ Pass ☐ Fail
2.	The shelf-to-column safety lock shall be tested at a minimum of ten random locations. Personnel shall attempt to separate empty shelf beams from the column without disabling the shelf-to-beam safety lock. If a shelf separates from a column then the system fails the test.	☐ Pass ☐ Fail
3.	Inspect each upright to see if each footplate is anchored. If any footplate is not anchored with a minimum 1/2-inch anchor bolt the system fails the test.	☐ Pass ☐ Fail

